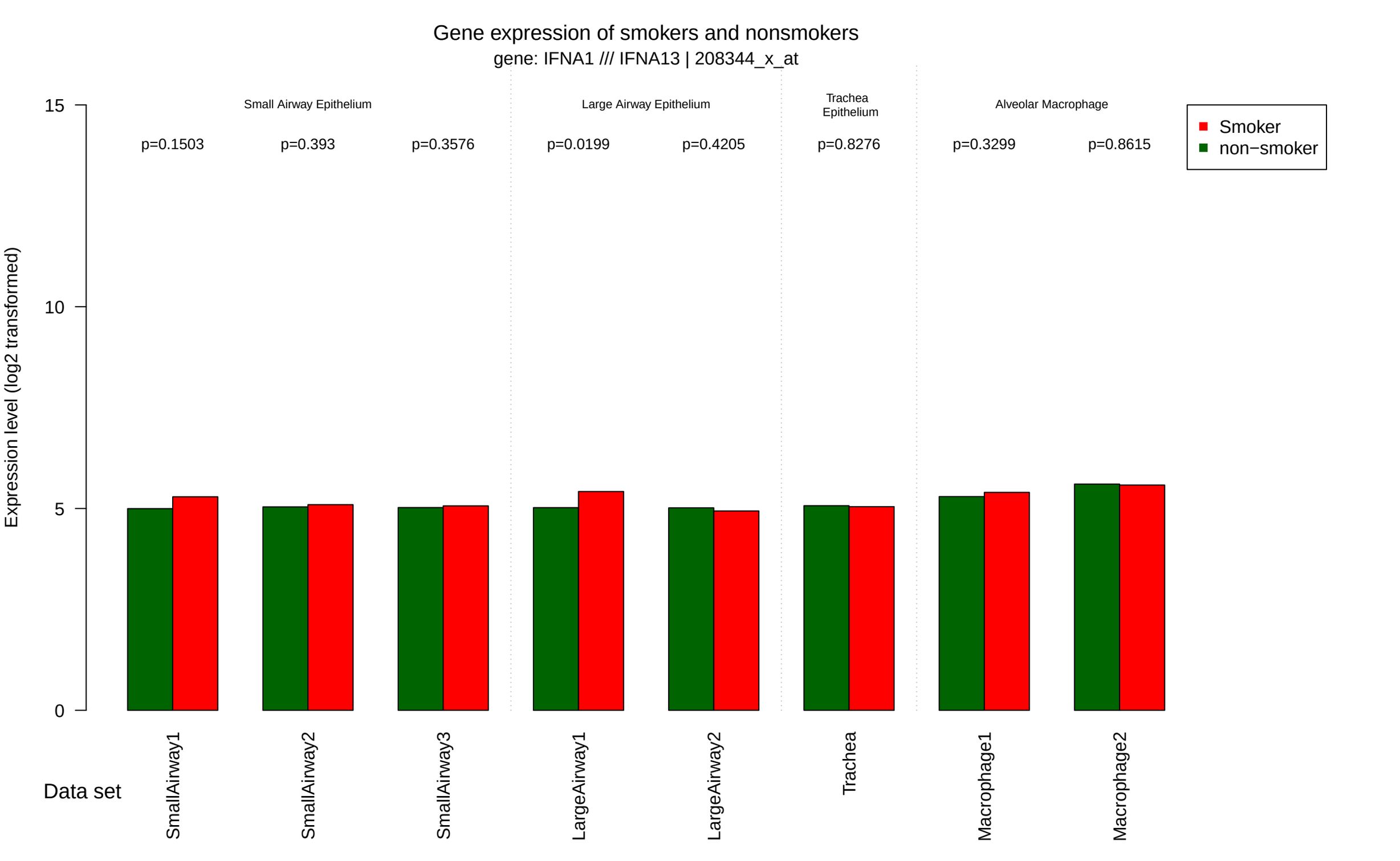
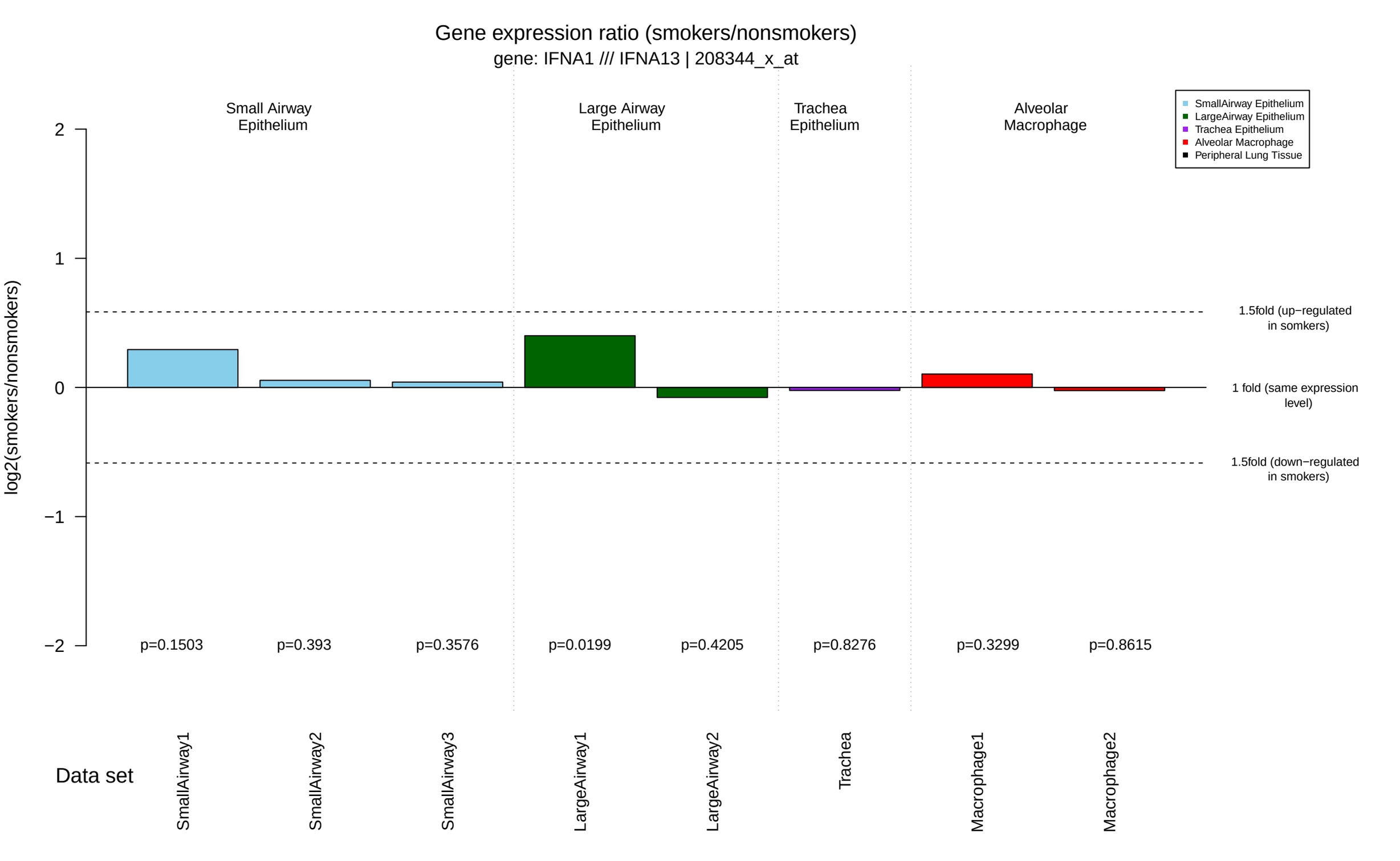
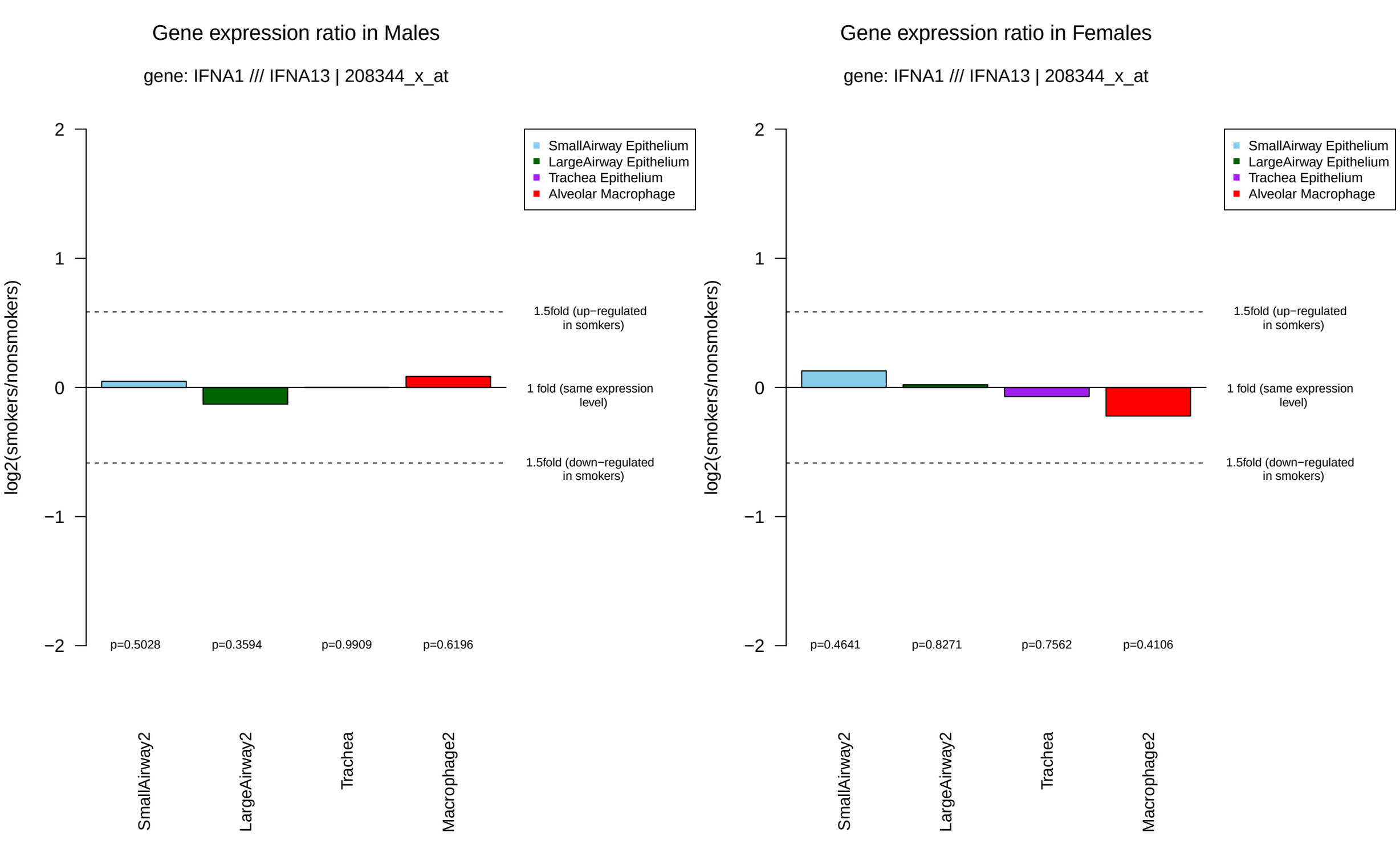
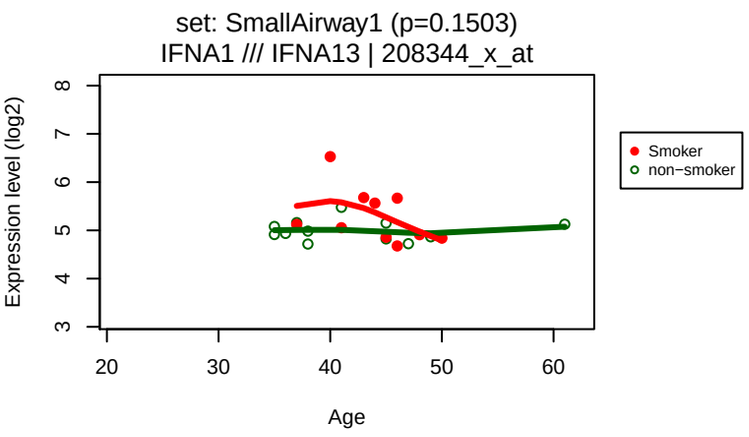
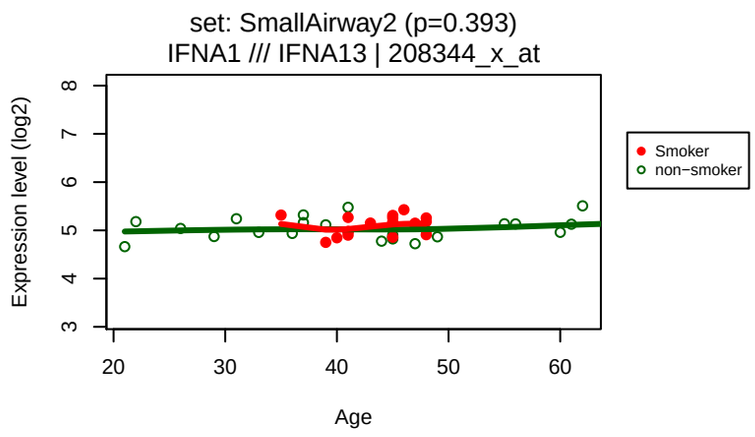
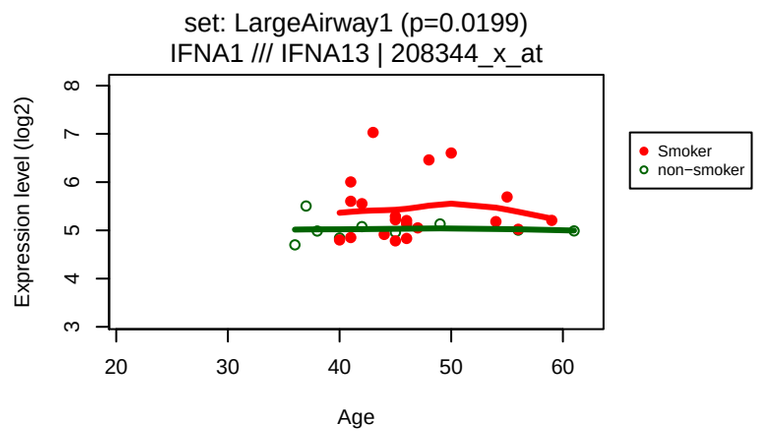

Gene expression of smokers and nonsmokers
gene: IFNA1 /// IFNA13 | 208344_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1503
p=0.393
p=0.3576
p=0.0199
p=0.4205
p=0.8276
p=0.3299
p=0.8615
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

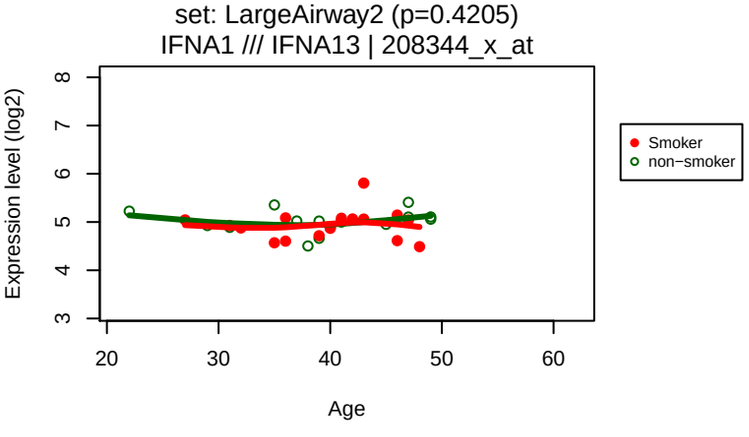

Gene expression ratio (smokers/nonsmokers)
gene: IFNA1 /// IFNA13 | 208344_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1503
p=0.393
p=0.3576
p=0.0199
p=0.4205
p=0.8276
p=0.3299
p=0.8615
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

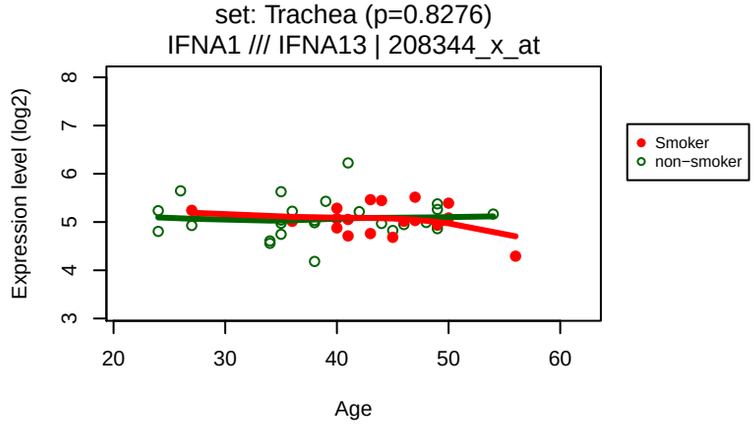

Gene expression ratio in Males
gene: IFNA1 /// IFNA13 | 208344_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5028
p=0.3594
p=0.9909
p=0.6196
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IFNA1 /// IFNA13 | 208344_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4641
p=0.8271
p=0.7562
p=0.4106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

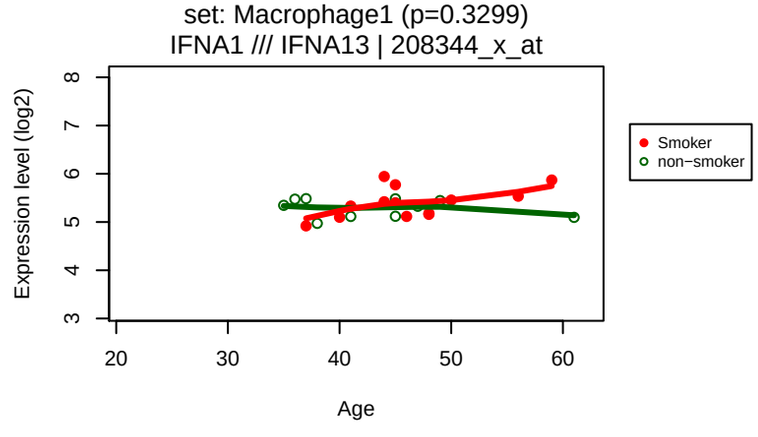

set: SmallAirway1 (p=0.1503)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker

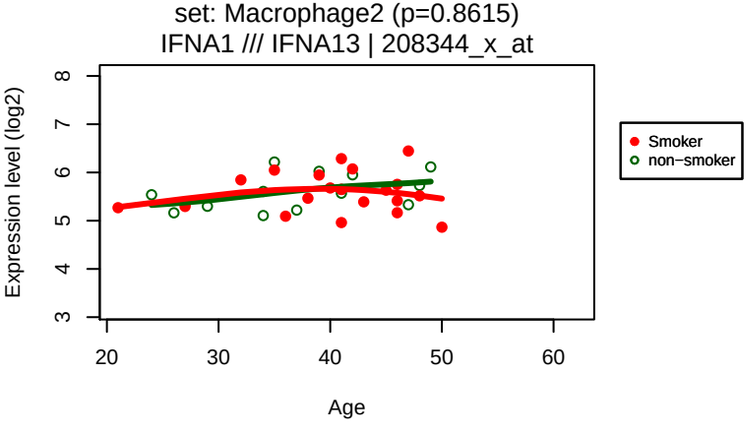

set: SmallAirway2 (p=0.393)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker

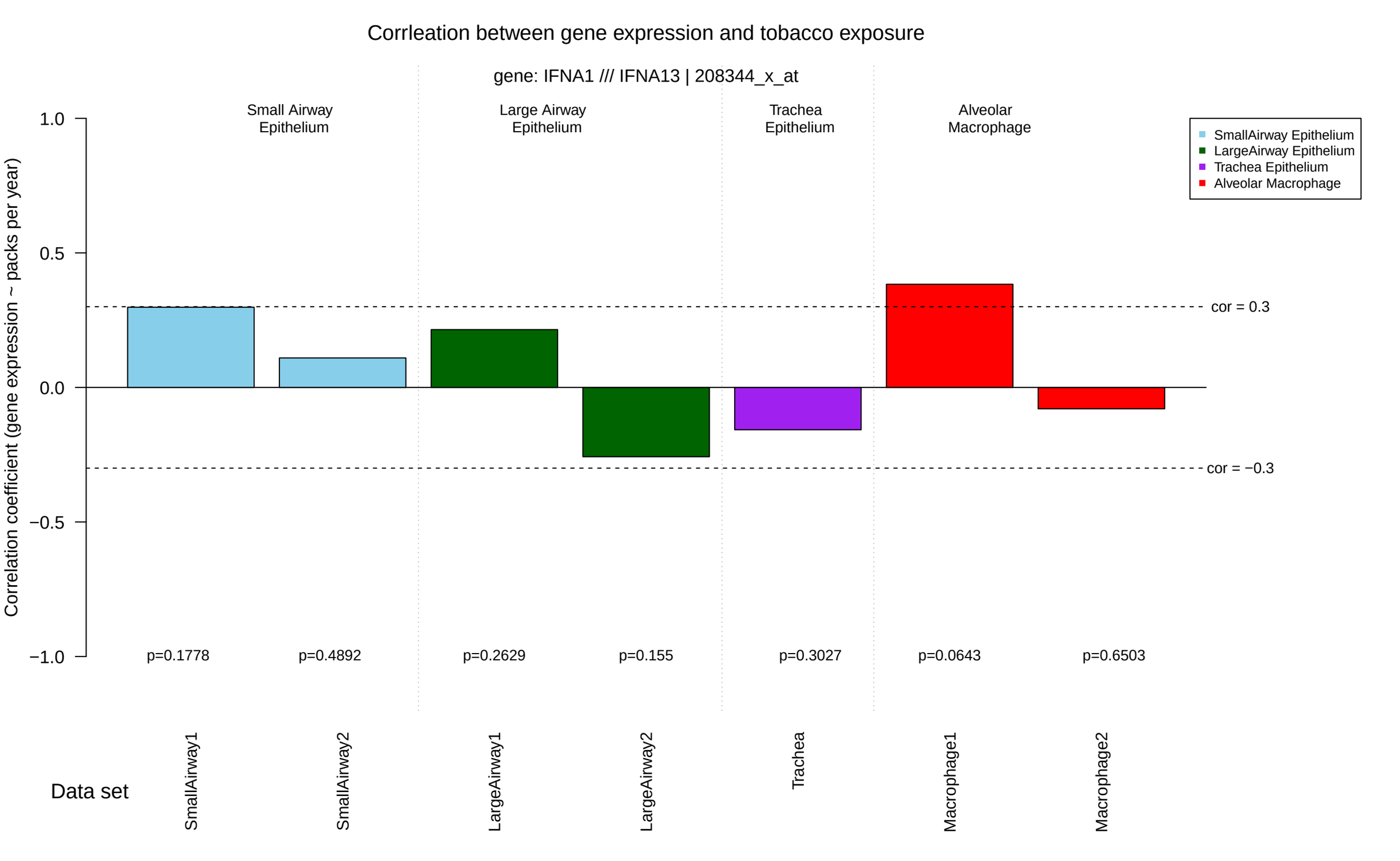

set: LargeAirway1 (p=0.0199)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4205)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8276)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3299)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8615)
IFNA1 /// IFNA13 | 208344_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IFNA1 /// IFNA13 | 208344_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1778
p=0.4892
p=0.2629
p=0.155
p=0.3027
p=0.0643
p=0.6503
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2